

Chapter 76 — Diagnosis and Management of Peyronie Disease

DEFINITION AND HISTORICAL PERSPECTIVE

- **American Urological Association (AUA) 2015 Guideline definition:** Peyronie disease (PD) is an acquired penile abnormality characterised by **fibrosis of the tunica albuginea**, which may be accompanied by pain, deformity, erectile dysfunction (ED) and/or distress.
- Originally termed *induratio penis plastica*. The eponym honours **François Gigot de la Peyronie (1743)**, the first to describe the condition and offer treatment.
- A historical review found possible earlier descriptions only by **Guilielmus de Saliceto (13th century)** and **Gabriele Falloppio (16th century)**; no other presumed precursor properly described or treated genuine PD.

GENERAL CONSIDERATIONS

- PD is an **acquired wound-healing disorder** of the tunica albuginea. Injury activates an abnormal healing response, which produces an inelastic scar (the "plaque").
- The resulting deformity may be **curvature, indentation, hinge effect or shortening**, and **ED is frequently associated**.
- **Cardinal feature: the scar does not undergo normal remodelling**, so the scar and the deformity persist.
- Because the pathophysiology is incompletely understood, there is **no preventive strategy and no reliable way to reverse the scar**.
- One theory for the middle-age predominance is that the **ageing tunica is more easily injured** in susceptible individuals.

BOX 76.1 PEYRONIE DISEASE CAVEATS — MISCONCEPTIONS CORRECTED

- **PD is not rare.** Prevalence is 3–20%, and higher in diabetes mellitus and ED.
- **Spontaneous resolution is rare.** Up to half of patients worsen if untreated in the first 12–18 months; the advice to "wait 6–12 months before treating" is a myth.
- **Not only middle-aged men:** PD occurs from the teens to the late 70s.
- **Not only White men.**
- **Trauma to the flaccid as well as the erect penis** can activate scarring in susceptible individuals.
- **ED is common** in PD.
- **Calcification is not a marker** of mature, chronic-phase disease.
- There is an **active phase** of scar growth, progressive deformity and pain.
- **Once stabilised, PD usually does not progress further**, and pain typically subsides.

ANATOMIC CONSIDERATIONS

- The **tunica albuginea** is multilayered and composed mainly of **type I collagen**, arranged as an **inner circular and an outer longitudinal layer** interlaced with elastin, with an incomplete septum anchored into the inner circular layer. It measures **1.5–3.0 mm** depending on position.

- **The septum is essential to structural integrity.** Computer models show that a single contiguous corporal body at full erection would generate enough stress to rupture the tunica.
- **Septal anchor sites are prone to microvascular trauma and delamination,** a proposed trigger for PD. Intracavernous pillars further reinforce the structure, with finer pillars at 5 and 7 o'clock (Fig. 76.2).
- **60–70% of plaques are dorsal,** usually with a septal association. Coital pressure may delaminate the layers and trap the healing process within the tunica.
- The **longitudinal layer is thinnest at 3 and 9 o'clock and absent between 5 and 7 o'clock.** This favours dorsal buckling and explains why **dorsal curvature is the most common.**
- In normal tissue **the layers slide on one another,** giving flexibility, rigidity and strength despite the thin tunica.
- **Plaque histology (Fig. 76.3):** collagen fibrils are disorganised and non-polarised, and elastin is reduced and haphazard. This causes **asymmetric corporal expansion, with deviation toward the side of the plaque.** A **circumferential plaque produces an hourglass deformity.**
- **Pathogenetic cascade (Fig. 76.4):** transforming growth factor (TGF)- β 1, platelet-derived growth factor, fibroblast growth factor, interleukins, tumour necrosis factor, nuclear factor- κ B, inducible nitric oxide synthase, reactive oxygen species and plasminogen activator inhibitor.

Etiology/Pathophysiology

- The exact cause is undefined. The consensus is that an **injurious stimulus in a susceptible individual** triggers the cascade.

Trauma

- Trauma may be a **single event or repetitive microtrauma.** It was first proposed as the primary cause in 1957.
- **Mechanism:** erect intracorporeal pressure is high and rises further with external force during intercourse. This can exceed tunical elasticity and cause a **microfracture.** Repetitive microtrauma causes **delamination** of the tunical layers and the vessels between them, then **microhaemorrhage,** which initiates the wound-healing cascade.
- **Trauma to the flaccid penis** (motor vehicle accident, sport) can also trigger PD. Once the scar develops, an inflammatory response causes pain, both flaccid and on pressure.
- **Dorsal and ventral shear stresses** during coitus may explain the predominantly dorsal plaque location.
- An electron-microscopy study found that **plaques from PD and from delayed surgery for penile trauma share the same ultrastructure.** This supports cavernous body trauma as the initiator.
- **Prior invasive procedures and genital or perineal trauma** increase the risk of PD.
- **Only about 16–40% of patients recall antecedent trauma.** The most commonly recognised event is **injury during sexual activity.** Anecdotally, the partner-on-top position is most often implicated, through a "**faux pas de coït**" (missed thrust) that produces high intracorporeal pressure.
- **Trauma alone is insufficient.** In a large penile fracture series, **no patient developed PD.** Proposed cofactors are genetic predisposition, autoimmunity, aberrant local wound healing and infection.
- **Medicolegal caution:** do not attribute PD to ED treatment, flaccid trauma, catheterisation or endoscopy. These provide an opportunity for disease in the genetically susceptible individual rather than causing it.

Concept — PD is trauma meeting a susceptible host. Septal delamination triggers a scar that never remodels, and the scar's position dictates the deformity.

PRESENTATION — SYMPTOMS

- The commonest presenting features are **pain, erect deformity, palpable plaque and ED**. Many patients present having **self-diagnosed ED**. Not every patient has pain or a palpable plaque, but shortening, hinge, distal softening and curvature are readily recognised.
- **Pain:**
 - In the acute phase, pain occurs **flaccid on palpation, with erection or during intercourse**, and most resolves once the disease is stable.
 - Some patients keep "**torque**" pain, a pulling sensation on the plaque during a strong erection. It **must not be confused with acute inflammatory pain**.
- **Curvature and function:**
 - Many patients can have intercourse with **up to 60°** of curvature, especially dorsal, gradual curves. **Ventral and lateral curves make intromission harder**.
 - Partners commonly report no discomfort, whatever the degree or direction.
 - **Patient estimates of curvature are unreliable** and tend to be overestimates.
- **Kelami classification (1983): $\leq 30^\circ$ mild, 31–60° moderate, $>60^\circ$ severe**. Some patients have no curvature but an **hourglass deformity with an unstable erection**.
- **Plaque configurations:** cords, nodules, coin-like, dumbbell and I-beam.
 - **Virtually all plaques have a septal component**, which supports the delamination theory.
 - **Pure septal plaques** may cause narrowing or shortening, or no recognised deformity.
- **Plaque orientation defines the deformity:**
 - A simple dorsal plaque produces dorsal curvature.
 - Transverse or spiralling scars (partial or circumferential) produce indentation or hourglass deformity, leading to an **unstable penis or hinge effect**, because the penis cannot tolerate axial load (Fig. 76.9).
- **Distal softening beyond the plaque:** dynamic infusion cavernosometry and cavernosography (DICC) shows **equal pressures proximal and distal to the plaque**. The cause remains speculative: local cavernosal fibrosis or site-specific venous leak.

Active vs. Stable Disease

- **Full spontaneous resolution is extremely rare**; when it occurs, it is more likely within 12–18 months.
- **Up to 50% worsen untreated**.
- **Active phase:** scar growth, progressive deformity and pain. **Stable phase:** usually no further progression.

Differential Diagnosis

- PD must be distinguished from **congenital conditions**: chordee without hypospadias, congenital penile curvature, and epispadias with curvature.
- **Early PD may be only focal tunical or septal thickening**, hard to see even on ultrasound.
- **Deformity without an obvious plaque:** consider post-traumatic cavernosal fibrosis, chronic inflammation, and benign or malignant tumours. **Epithelioid sarcoma of the penis** is rare, presents as a focal lesion and can mimic PD.

Evaluation and Diagnosis

History

- **Symptoms:** pain, deformity and plaque; gradual or sudden onset; timing; and any **inciting event** — trauma to the flaccid or erect penis, or instrumentation.
- **Personal or family history of fibrotic disorders:** Dupuytren disease (DD) and Ledderhose disease (Fig. 76.5).
- **Key question:** can the patient achieve **intromission**, given the deformity and/or rigidity?
- **Predictive question:** *"If your penis was straight with the same quality of rigidity that you have now, would it be adequate for penetrative sex?"*
 - This is an **effective predictor of postoperative erectile function**.
 - If the answer is "no," with or without medication, direct the patient toward a **penile prosthesis with straightening**. Other approaches may straighten the penis but will not give a functional erection.
- **Also ask about: vascular risk factors** for ED (diabetes, hypertension, hypercholesterolaemia and smoking), premature or delayed ejaculation, and medications, which may reveal predisposing conditions.
- **Peyronie's Disease Questionnaire (PDQ): 15 items in 3 domains** — psychological and physical symptoms (6 items), penile pain (3 items) and symptom bother (6 items). Each domain is scored independently, with **no total score**; higher scores mean greater negative impact.
- **Home erect photograph:** controversial, because a two-dimensional image cannot capture a three-dimensional deformity. Smartphone photographs from above and from the side give a useful general impression.

Physical Examination

- **General:** femoral pulses, appearance of the flaccid penis, circumcision status.
- **Examine the penis on stretch** to identify the plaque (Fig. 76.6).
- Record plaque location, but **size measurement is inaccurate by any modality**, because the plaque is rarely discrete, has irregular borders and extends into a septal cord. **A reduction in plaque size after treatment does not correlate with improved deformity.**
- **Stretched penile length (SPL)** is a critical baseline, since patients overestimate their length loss.
 - Grasp the glans and **stretch at 90° from the body**.
 - **Measure dorsally from pubis to corona** — two fixed points that make the measurement repeatable.
- **Consistency:** a **"rock-hard" plaque suggests calcification**; confirm by imaging, preferably ultrasound.

Calcification.

- On ultrasound, calcification is **hyperechoic with acoustic shadowing** (Fig. 76.7). Computed tomography and magnetic resonance imaging add little.
- **Calcification can occur early** and is **not** a marker of chronic, severe or mature disease. It likely reflects a **genetic subtype with osteoblastic gene activation**.
- **Extensive calcification predicts:**
 - **Poorer response to non-surgical therapy**, because intralesional verapamil and interferon (IFN) cannot penetrate mineralised tissue.
 - **A greater likelihood of needing a prosthesis.**

LEVINE CALCIFICATION GRADING (2013)

Grade	Definition	Surgical likelihood
1	<0.3 cm	Not increased
2	0.3–1.5 cm	Not increased
3	>1.5 cm in any dimension, or multiple plaques ≥1.0 cm	More likely to undergo surgery (when erectile function is satisfactory)

Intracavernosal Injection

- Specialist centres routinely perform duplex ultrasound with intracavernosal injection (ICI), especially in surgical candidates.
- **AUA 2015 (Expert Opinion):** clinicians should perform an in-office ICI test, with or without duplex Doppler ultrasound, **before any invasive intervention**. History and examination may suffice for diagnosis, but **ICI is required before invasive therapy**.

Duplex Doppler Ultrasonography and Other Objective Tests**BOX 76.3 VALUE OF PENILE DUPLEX ULTRASONOGRAPHY FOR PEYRONIE DISEASE**

- **Identifies and measures calcification.**
 - **Identifies corporeal fibrosis.**
 - Shows the **erectile response to vasoactive ICI**, which can be compared with the home erection.
 - **Measures vascular parameters:** peak systolic velocity, end-diastolic velocity and resistive index (RI).
 - Gives the **optimal objective measure of erect deformity:** curvature, girth irregularity and hinge.
- Penile duplex Doppler ultrasonography (PDDUS) findings are **critical for choosing the surgical technique** (Fig. 76.10). In early disease, PDDUS can detect **non-palpable septal plaques** that cause distal rigidity loss, length loss or pain without deformity.
 - Curvature is measured with a **goniometer** (Fig. 76.8).
 - **Preoperative erectile function correlates strongly with postoperative results.**
 - **Ventral curvature is most associated with cavernous veno-occlusive dysfunction**, which supports concern about postoperative ED after grafting ventral curves.
 - **DICC** can assess venous leak but is rarely performed. It adds invasiveness and cost for little gain over a well-done PDDUS.
 - **Sensation:**
 - There is no standard test; light touch and **biothesiometry** can be used, and the **AUA advises biothesiometry when sensory change is present**, to establish a baseline.
 - Its validity is controversial. Vibration sense is assumed to be a surrogate for sexual sensation, with the index fingers as positive control and the anterior thighs as negative control.
 - **Morning serum total testosterone** is recommended in **PD with ED** (per the AUA ED guideline), **not in PD without ED**.

KEY POINTS — EVALUATION

- **History:** symptom onset, vascular risk factors, patient-estimated deformity and rigidity.
- **Questionnaire:** the PDQ documents the effect of PD.
- **Examination:** plaque palpated on stretch, stretched flaccid length, and pain on palpation.

- **Objective erect deformity after ICI, especially if surgery is contemplated.**

TREATMENT

- Algorithms are guides only. **Management is individualised** to the history, examination, PDDUS and patient goals.

Medical Therapies

- Older studies were weak: small, uncontrolled, non-randomised and with few objective measures.
- **AUA 2015: all oral and topical agents except non-steroidal anti-inflammatory drugs (NSAIDs) are "should not offer."**
- **Reassurance alone** may suffice if there is no pain or difficulty with penetration. Reassure the patient that PD is **not cancerous and not life-threatening**.
- **Natural history under conservative management:**
 - Shortening is common and improvement uncommon. Pain usually resolves, but **a notable proportion have pain years later, including new-onset pain**, possibly from repeated trauma to the curved penis.
 - Another long-term survey found curvature improved, stabilised or worsened in varying proportions. **Erectile function and satisfaction declined, psychosocial distress was high, and traction users did better.**

Oral Non-Steroidal Anti-Inflammatory Drugs

- **AUA Expert Opinion:** may be offered for **pain in active PD**. Assess pain on a visual analogue scale (VAS) and reassess periodically.

Pentoxifylline

- A **methylxanthine** used for claudication. It reduces blood viscosity, improves red-cell deformability and reduces platelet aggregation.
- **Antifibrotic actions:** blocks the **TGF- β 1 pathway**, prevents type I collagen deposition, acts as a non-specific phosphodiesterase inhibitor, increases fibrinolysis, and inhibits elastogenesis via an α 1-antitrypsin-related mechanism. It reduced plaque in an animal model.
- Human evidence is **small, retrospective and confounded by concurrent traction**. **AUA: efficacy uncertain.**

Phosphodiesterase Type 5 Inhibitors

- **Phosphodiesterase type 5 inhibitors (PDE5Is) are safe and effective for ED in PD.**
- **Tadalafil plus extracorporeal shock wave therapy (ESWT)** improved International Index of Erectile Function (IIEF) and quality-of-life scores, but gave **no deformity benefit**. **AUA: ESWT should not be used to reduce curvature or plaque size.**
- **Antifibrotic rationale:** raised cyclic guanosine monophosphate (cGMP) inhibits collagen synthesis and induces fibroblast and myofibroblast apoptosis. **Low-dose daily tadalafil remodelled isolated septal scars.**
- **AUA: PDE5Is are recommended for ED but not for PD itself.**

Intralesional Injections

Collagenase Clostridium histolyticum (Xiaflex).

- **The first Food and Drug Administration (FDA)-approved drug for PD (2013).** It selectively degrades **collagen types I and III**, even in the presence of tissue inhibitors of metalloproteinases (TIMPs), which are elevated in PD. It spares elastic fibres, vascular smooth muscle and axonal sheaths.

- In early trials, **response was inversely related to curvature severity and plaque size**, and most patients had reduced plaque size.
- **Modelling is essential**: collagenase *Clostridium histolyticum* (CCH) plus modelling outperformed both placebo and CCH without modelling.
- **IMPRESS I and II — landmark phase III trials** (Investigation for Maximal Peyronie's Reduction Efficacy and Safety Studies):
 - **Protocol**: up to **4 cycles, 6 weeks apart** → each cycle is **2 injections of 0.58 mg**, 24–72 hours apart → **modelling 24–72 hours after the second injection**. Maximum **8 injections**.
 - Randomised **2:1** CCH to placebo, stratified by curvature 30–60° vs 61–90°.
 - **Result**: **34% improvement (–17°) vs 18.2% (–9.3°) with placebo**. The PDQ symptom bother score was significantly better.
 - **Excluded**: extensive calcification, **ventral plaques**, and disease duration <12 months.
 - Anti-CCH antibodies developed in virtually all patients, **without resulting adverse events**.
 - **Adverse events**: ecchymosis and bruising were commonest. Serious events were **corporal rupture** and **penile haematoma**, managed surgically or by percutaneous drainage.
- **Most serious complication: penile (corporal) fracture** (0–4.9%), because the injection weakens the tunica around the plaque. Pooled data show that most adverse effects are minor and serious events are rare.
- **Protocol modifications**: most of the gain occurs in the **first 3 cycles**, with no significant gain in the 4th. Shortened regimens gave IMPRESS-equivalent results. There is **no consensus protocol**, and **cost** is a major concern.
- **Acute-phase use**: improvement was comparable to stable disease, but **FDA approval is for stable disease** and more data are needed.
- **Subsequent plication or grafting is not more difficult** after CCH.
- **Long-term (up to 5 years)**: no safety signals, and curvature and PDQ improvements were maintained without further injections. Results were similar when calcification did not interfere with injection.
- **Real-world multicentre data**: about 30% curvature reduction, with more cycles giving greater benefit.
- **Ventral and lateral curves**: relative improvement exceeded dorsal curves, and no urethral complications were seen in a small series. CCH is **not yet routine for ventral curves**, because the corpus spongiosum and urethral submucosa also contain type I and III collagen.
- Since approval, CCH is increasingly the **first-line** choice.
- **Predictors of poor response**: calcification (including grade 3), curvature ≥60°, severe indentation or narrowing, and hinge. **Pretreatment IIEF-5 ≥17 is associated with a better response**.
- **AUA**: CCH for **stable curvature >30° and <90° with intact erectile function**, with or without medication; counsel on risks.

Interferon Alfa-2b.

- **Mechanism (in vitro)**: dose-dependent reduction in plaque fibroblast proliferation, reduced extracellular collagen and increased collagenase.
- **The first multicentre, randomised, placebo-controlled trial of intralesional therapy for PD** (single-blind) found interferon alfa-2b (IFN α-2b) significantly better than saline for curvature, plaque size and density, and pain. **Clinical significance was questioned** given cost and side effects. It also showed that **saline injection has little or no effect**.
- **Side effects**: flu-like symptoms (fever, chills, arthralgia), sinusitis, minor penile swelling and ecchymosis. **Pre-injection NSAIDs** are effective, and effects last <36 hours.
- **IFN versus verapamil**: similar outcomes, but **verapamil is far cheaper**.

- **AUA:** may administer IFN α -2b, with counselling on adverse events. It has had **limited US availability since 2022**, which has increased verapamil use.

Verapamil.

- **Mechanism:** calcium channel blockade inhibits proline incorporation into extracellular matrix (ECM), reduces fibroblast proliferation and ECM synthesis and secretion, and increases collagenase activity. In a rat model, collagen, elastin and alpha-smooth muscle actin (ASMA, a myofibroblast marker) all decreased.
- Verapamil was the **first new intralesional agent since steroid injection**.
- Uncontrolled series showed improvement in pain, narrowing, distal rigidity and curvature.
- **Controlled trials are conflicting:** one single-blind trial showed improved plaque size, narrowing and erection quality but **no curvature benefit**, and one placebo-controlled trial showed **no difference from saline**. At a minimum, verapamil can **stabilise** deformity.
- **Dilution:** 10 mg/10 mL is standard, but **greater dilution (10 mg in 20 mL) gave a better response**.
- **Poor candidates:** extensive calcification, curvature $>90^\circ$, and **ventral curvature**, where adequate infiltration is difficult.
- **Predictors of success:** younger age (<40 years) and curvature $>30^\circ$.
- **Randomised head-to-head trial against CCH** (non-blinded, same modelling protocol): **CCH gave greater curvature improvement**, more complete resolutions, and better PDQ and IIEF-5 scores, but more minor adverse events (bruising, swelling, pain).
- **AUA:** may offer verapamil, with counselling on injection-site effects.

TABLE 76.2 INTRALESIONAL AGENTS FOR PEYRONIE DISEASE — REBUILT, WITH AUA 2015 POSITION

Agent	Mechanism	Outcome	Adverse effects	AUA 2015
Verapamil	Calcium channel blocker: $\downarrow$ fibroblast proliferation, $\downarrow$ ECM synthesis and secretion, $\uparrow$ collagenase	$\downarrow$ curvature and narrowing, $\uparrow$ erection quality (inconsistent in controlled trials)	Nausea, light-headedness, penile pain, ecchymosis	May offer
Nicardipine	Dihydropyridine calcium channel blocker; in vitro more effective than verapamil at $\downarrow$ glycosaminoglycan and ECM	$\downarrow$ pain, $\uparrow$ IIEF-5, $\downarrow$ plaque; no curvature benefit	No severe effects (no hypotension or cardiovascular events)	—
IFN α-2b	$\downarrow$ fibroblast proliferation (dose-dependent), $\downarrow$ collagen, $\uparrow$ collagenase	Curvature $\downarrow$ 27% (13.5°) vs 9% (4.5°) placebo	Sinusitis, flu-like symptoms, swelling, ecchymosis	May administer
CCH	Degrades collagen I and III despite TIMPs	Curvature $\downarrow$ 34% (17°) vs 18% (9.3°) placebo; PDQ bother improved	Contusion, ecchymosis, corporal rupture	Stable, $>30^\circ$ and $<90^\circ$, intact erections

- **What the table tests:** only CCH is FDA-approved and supported by phase III placebo-controlled data. **Nicardipine improves symptoms and plaque but not curvature**, and verapamil is cheap but inconsistent.

Penile Traction Therapy

- **Indication:** a **non-invasive, non-surgical first option**.
- **Mechanism — mechanotransduction:** extrapolated from DD, controlled stretch remodels collagen, reduces myofibroblast activity and increases matrix metalloproteinases (MMPs). Tension **reorients collagen parallel to the axis of stress**. In PD cells in vitro, **ASMA decreased and collagen-degrading MMPs increased**, without upregulation of heat shock protein 47 or the TGF- β 1 receptor.
- **None of the classic clinical studies were controlled.**

- **Pilot data** (2–8 hours a day for 6 months): curvature, SPL, girth and hinge all improved, and results were maintained at 6 months after stopping, with no adverse events.
- **In stable disease with curves $\leq 50^\circ$** , curvature change was minimal, but satisfaction was reasonable, **mainly from length gain**. Selection of stable, often calcified disease may have underestimated efficacy.
- **Acute phase** (progressive curvature $>15^\circ$ and/or pain in the last 12 months):
 - **Regimen:** 6–9 hours a day, not during sleep, **removed for 30 minutes every 2 hours to prevent glans ischaemia**.
 - **Results:** curvature, pain and penetration ability improved, and untreated controls worsened. **The need for surgery fell, and surgery was simplified from grafting to plication in some.**
 - **Adverse effects:** balanopreputial erythema and discomfort; no sensory change.
- **The barrier to adoption is long daily wear time.** RestoreX, a counter-bending device worn **30–90 minutes a day**, improved length, curvature and IIEF-5 in a randomised controlled trial of a difficult cohort (long-standing disease; hourglass and complex curves not excluded). Most patients preferred it as first line.
- **AUA 2015 makes no recommendation on penile traction therapy (PTT)**, although adoption is increasing.

Vacuum Therapy

- **Rationale:** indirect, repeated stretch reorganises ECM, analogous to the management of DD.
- **10 minutes twice daily** improved curvature compared with non-users, with the greatest gain in larger curves, **independent of age and disease duration**. Short stretch durations are **unlikely to produce true mechanotransduction**.
- **Complications** (with constriction rings or excessive pressure): PD itself, urethral bleeding, skin necrosis and ecchymosis.
- **AUA: no recommendation.** A vacuum erection device (VED) is an option for patients who choose conservative management.

Combination Therapy

- **PTT plus verapamil plus pentoxifylline plus L-arginine:** both arms improved, with no difference in curvature. **Traction preserved or gained SPL, while non-users lost length.** Benefit showed a **dose-response at ≥ 3 hours a day**.
- **Verapamil (injection and iontophoresis) plus a multi-agent supplement pill:** greater reduction in plaque and curvature.
- **PTT plus IFN:** SPL gain without curvature change, **only with ≥ 3 hours a day**.
- **AUA: no recommendation.**

TABLE 76.1 ORAL AGENTS FOR PEYRONIE DISEASE (PD)

Agent	Mechanism	Outcome	Adverse effects
Potaba	$\uparrow$ monoamine oxidase activity $\rightarrow$ $\downarrow$ serotonin $\rightarrow$ antifibrotic	$\downarrow$ plaque size, no curvature change	Anorexia, nausea, fever, rash, hypoglycaemia, acute hepatitis
Vitamin E	Antioxidant ($\downarrow$ reactive oxygen species)	No benefit	Possible cerebrovascular events, gastrointestinal upset, headache, dizziness
Tamoxifen	Induces TGF- β (oestrogen receptor-independent); macrophage deactivation, T-cell suppression	No benefit	Alopecia, retinopathy, thromboembolism, pancytopenia
Colchicine	Microtubule depolymerisation; $\downarrow$ mitosis and leucocyte adhesion; $\uparrow$ collagenase	No benefit	Myelosuppression, gastrointestinal upset

Agent	Mechanism	Outcome	Adverse effects
Carnitine	↑ mitochondrial respiration, ↓ free radicals	No benefit	Seizures, gastrointestinal upset
Pentoxifylline	Blocks TGF-β1; ↓ type I collagen; non-specific phosphodiesterase inhibitor; ↓ platelet-activating factor	Curvature ↓ in a minority	Nausea, vomiting, dyspepsia, malaise, flushing, dizziness, headache

- **What the table tests: no oral agent reliably reduces curvature.** Potaba shrinks plaque without straightening the penis, and pentoxifylline has only weak evidence. This is why **the AUA endorses only NSAIDs, for pain.**

TABLE 76.3 EXTERNAL FORCE APPLICATION FOR PEYRONIE DISEASE

Modality	Mechanism	Outcome	Adverse effects
Electromotive drug administration	Bypasses hepatic metabolism, ↑ tissue drug level	Verapamil alone: no benefit. With dexamethasone: ↓ plaque and curvature	Transient erythema at the electrode
ESWT	Direct plaque injury → inflammation → macrophage lysis	↓ pain, ↑ IIEF-5 and quality of life; no curvature benefit	Petechiae, ecchymosis
Traction	↓ ASMA, ↑ MMPs	↑ length and girth, ↓ curvature and pain, plaque softening	Balanopreputial erythema, discomfort
Vacuum	Unknown; traction-like	Modest curvature ↓	PD, urethral bleeding, necrosis, ecchymosis
Radiotherapy	Anti-inflammatory (leucocyte adhesion and nitric oxide synthase modulation)	No benefit	Possible malignant change, ↑ ED risk in the elderly

- **What the table tests: only sustained mechanical stretch (traction) changes the deformity.** ESWT treats pain only, and radiotherapy has no benefit but carries real risk.

Conclusion — Non-Surgical Therapy

- The **minimum goal** is to prevent progression during the acute phase. The **ultimate goal** is to reduce deformity and restore sexual function.

KEY POINTS — NON-SURGICAL TREATMENT OF PD

- Patients with no pain or difficulty with penetration may need **reassurance only**.
- Conservative treatments give inconsistent or modest improvement, because the underlying mechanism is poorly understood.
- **No oral agent** shows clinically meaningful curvature benefit in placebo-controlled trials. **NSAIDs for pain are the only AUA-endorsed oral therapy.**
- **Topical therapy and ESWT do not reduce deformity.**
- **Verapamil and IFN α-2b** reduce curvature and improve function in mostly uncontrolled studies, and appear to **stabilise deformity in the acute phase.**
- **CCH is the only FDA-approved agent:** 34% (17°) vs 18.2% (9.3°) with placebo in phase III.
- **Counter-bending traction devices (RestoreX)** with short wear times improve curvature, length and erectile function, even as monotherapy.

SURGICAL MANAGEMENT

- Practice varies widely internationally. **Few surgeons are high-volume** (defined as >20 plications a year), and most perform fewer than 10 graft procedures annually.

Indications

- **Surgery is the gold standard** for the most rapid and reliable correction of deformity. A **penile prosthesis is added when ED coexists**.
- Deformity that precludes intercourse, causes pain to either partner, or causes **distress at the appearance of the erect penis** is an indication.

BOX 76.4 INDICATIONS FOR SURGERY

- **Stable disease: ≥ 12 months from onset, with the deformity stable for ≥ 6 months.** This is expert consensus and has not been formally studied.
- **Inability to have satisfactory penetrative intercourse** because of deformity and/or inadequate rigidity.
- **Failed conservative treatment.**
- **Desire for the most rapid and reliable result.**

- **No single operation is the standard of care.** The choice depends on severity and direction of curvature, hinge, erection quality and patient goals (Fig. 76.10).

Preoperative Consent

- Patients are often emotionally devastated. **Dissatisfaction often stems from expecting a return to the pre-PD appearance**, so shared decision-making and expectation-setting are essential.

BOX 76.5 PREOPERATIVE CONSENT

- **Persistent or recurrent curvature:**
 - Uncommon (up to 16%), and rarely needs reoperation.
 - The goal is "**functionally straight**": $\leq 20^\circ$ (expert opinion). The **European Association of Urology (EAU)** defines success as $\leq 15^\circ$.
- **Length:**
 - Some length loss occurs with every procedure, and it is **more likely with plication than grafting**.
 - **70–80% have already lost length from the disease**, so document SPL preoperatively.
- **Diminished rigidity:**
 - About 5% in all studies, grafting more than plication; about 30% if preoperative rigidity is suboptimal. Up to 50% have some reduction, which may respond to a PDE5I.
 - **Preoperative erection quality is the most reliable predictor of postoperative ED.**
 - Straightening does not usually improve rigidity, although correcting geometry improves it in some.
 - Refractory ED can be treated with a PDE5I, ICI, a VED or a later prosthesis, placed with little added difficulty.
- **Decreased sensation:** about 20% report some reduction, which **rarely compromises orgasm or ejaculation**. Acute hyperaesthesia or hypoaesthesia usually settles within months, and stabilises by 6–12 months.

CHOICE OF PROCEDURE — THE TWO DETERMINANTS: PREOPERATIVE RIGIDITY AND SEVERITY OF DEFORMITY

Scenario	Procedure
Adequate rigidity ($\pm$ PDE5I), simple curve $<60\text{--}70^\circ$, no hourglass or hinge, expected length loss $<20\%$ of erect length	Tunical plication
Strong rigidity; complex curve $>60\text{--}70^\circ$, destabilising hourglass or hinge, or short phallus (<9 cm stretched) ; patient accepts ED risk	Plaque incision or partial excision with grafting
ED refractory to medical therapy	Penile prosthesis $\pm$ modelling, plication, incision or graft

- Expected length loss is estimated on the erect penis as the **difference between the long and short sides**.

FIG. 76.10 ALGORITHM FOR THE SURGICAL MANAGEMENT OF PEYRONIE DISEASE

1. **Is the erection adequate for penetration ($\pm$ PDE5I)?**
2. **No** → **penile prosthesis** (or pre-prosthesis plication). If not functionally straight → **manual modelling**. If still not straight → **plaque incision, with a graft if the defect is >2 cm**.
3. **Yes** → duplex ultrasound of the deformity:
 - **Simple uniplanar curve $<60\text{--}70^\circ$, no hourglass or hinge** → **plication**, if the patient accepts the risk of length loss. If not → **plaque incision or partial excision and grafting**.
 - **Complex or multiplanar curve $>60\text{--}70^\circ$, severe indentation or hourglass, or short phallus (<9 cm)** → **plaque incision or partial excision and grafting**, if the patient accepts the risk of ED. If not → **penile prosthesis**.

Tunical Shortening Procedures**Plication and Corporoplasty**

- **Principle:** shorten the long (convex) side to match the short side, always **on the side opposite the curvature**.
- **Advantages:** short operating time, good cosmesis, minimal effect on rigidity, simplicity, safety and effective straightening.
- **Disadvantages:**
 - It **shortens the penis** and does not correct existing shortening.
 - It **does not correct, and may worsen, hourglass or hinge**, leaving an unstable penis.
 - The plaque is left in situ.
 - Narrowing or indentation, knot pain and suture granulomas can occur.
- **Nesbit failure predictors:** impaired preoperative erectile function, shortening >2 cm, and curvature $>30^\circ$.

COMPARISON — PLICATION TECHNIQUES (FIGS. 76.11–76.14)

Technique	Incision	Method	Key point
Nesbit (1965)	Full-thickness transverse elliptical excision of tunica	Wedges excised contralateral to the curve; closed transversely with permanent suture ($\pm$ absorbable)	The original. For a ventral curve, elevate Buck fascia and excise dorsal wedges
Yachia (1990)	Full-thickness vertical incision, 0.5–1.5 cm; no tunica removed	Heineke-Mikulicz: incise vertically, close transversely	Keep the incision short, or transverse closure narrows and destabilises the shaft. Claimed less perceived shortening

Technique	Incision	Method	Key point
Essed-Schroeder (1985)	None	Non-absorbable figure-of-eight sutures with buried knots	The first plication without incision
Ebbehoj-Metz (1987)	None	Multiple rows of plicating sutures	Described for congenital curvature
16-dot	None	Permanent sutures in an extended Lembert pattern, 4 dots per plication	Popular
Tunica albuginea plication (TAP) (Levine modification of Duckett-Baskin)	Partial thickness: paired transverse parallel incisions (1–1.5 cm long, 0.5–1.0 cm apart) through the longitudinal fibres only , sparing the inner circular layer	Excise the intervening longitudinal fibres to reduce bulk; single central permanent suture (2-0 braided polyester, Tevdek or TiCron) as a buried inverting vertical mattress , covered by 3-0 polydioxanone (PDS) Lembert sutures	Cavernosal tissue undisturbed, so a lower theoretical ED risk. Originally for congenital curvature in children

- **What the table tests:** the techniques differ in **incision depth** (full thickness, partial thickness or none) and in **whether tunica is removed**. **All shorten the long side**, so none corrects hourglass, hinge or existing shortening.
- **Length loss after TAP** depends on curve direction and degree, and is **greatest with ventral curves >60°**. Preoperative length and curve characteristics predict satisfaction.
- **Outcomes:**
 - Straightening 79–100%, satisfaction 65–100%, and recurrence >30° in up to 12%.
 - New ED 0–38%, diminished sensation 4–21%, haematoma up to 9%, urethral injury <2%, and phimosis up to 5%.
 - Long-term TAP data show durable correction with few complications at 10 years.
- **International Consultation on Sexual Medicine (ICSM, 2010):** **no approach is superior**, and plication corrects curvature with **less new ED than grafting**.
- **AUA:** may offer plication if rigidity is adequate for coitus ($\pm$ pharmacotherapy or VED). **Plication is the most common PD operation**, accounting for about half of all PD surgery.
- **Table 76.5 (Outcomes of Tunical Shortening Procedures — Nesbit, plication and Yachia series):** straightening, shortening, ED and satisfaction vary widely between series, and **shortening is reported frequently**, especially after Nesbit and plication.

Tunical Lengthening Procedures

- **Indications for plaque incision and grafting (PIG) or partial plaque excision and grafting (PEG):** complex disease with some or all of **curvature ≥ 60 –70°**, **shaft narrowing**, **hinging** and **extensive calcification**.
- **The patient must have strong preoperative erections.** Ask: *"If your penis was straight, would your current rigidity allow penetrative sex?"* If he hesitates, grafting should not be performed unless he accepts the risk of worse ED and a possible later prosthesis.
- Some patients refuse a first-line prosthesis, and some plication candidates refuse because of fear of length loss. **Grafting can be offered to both groups**, since a prosthesis can be placed later with minimal added difficulty.
- **Grafting advantage:** it corrects curvature, restores shaft calibre, and may recover **0.5–3.0 cm of length**.
- **Predictors of postoperative ED** (from single-centre data): age >55 years, veno-occlusive dysfunction on PDDUS (RI <0.80), large defect or graft, **ventral curvature**, and curvature >60°.

- **The strongest criterion is preoperative erection quality; preoperative and postoperative ED are linearly associated.**
- **Ventral deformities do poorly with grafting**, because of associated veno-occlusive dysfunction.

Plaque Incision and Grafting

- **Total plaque excision has been abandoned**, because large grafts caused unacceptable ED.
- **Incision options:**
 - The **Gelbard modified-H or double-Y incision** at the point of maximum curvature expands the tunica, corrects curvature and calibre, and minimises cavernosal exposure. **Multiple small incisions with grafts** give a smoother correction.
 - **Multiple transverse incisions with a rhomboid defect** were the first description of PIG.
 - The **Egydio incision** (near-circumferential extension of the H) is associated with **more ED**.
- **Mechanism of postoperative ED:** cavernosal fibrosis and disruption of the veno-occlusive mechanism.
- **Systematic review: no incision type is superior**, and straightening is 80–96%. Many patients depend on erectile aids afterward, which exposes a weakness of the IIEF-5 (Sexual Health Inventory for Men): it does not control for aid use. Sensory change is mostly transient.

Plaque Excision and Grafting

- **Preferred when the area of maximum deformity must be excised**, as with **severe indentation**. Otherwise the straightened penis may remain narrowed and unstable.
- **Dart the corners of the defect radially** to correct narrowing.
- **Egydio's geometric principles** size the graft, but are considered unnecessarily complex and carry higher ED rates.
- **Key to reducing ED: minimise trauma to the underlying cavernosal tissue**, preserving the veno-occlusive relationship between cavernosal tissue and graft.
- **AUA:** may offer PIG/PEG if rigidity is adequate for coitus ($\pm$ pharmacotherapy or VED) (Moderate Recommendation; Evidence Grade C).

Graft Materials

- **The ideal graft** approximates the strength and elasticity of tunica, has minimal morbidity and tissue reaction, is readily available, thin, pliable, easy to size and suture, cheap and infection-resistant, and preserves erectile capacity.

COMPARISON — GRAFT MATERIALS

Category	Examples	Status and issues
Autologous	Dermis, fat, tunica vaginalis, dura mater, temporalis fascia, saphenous vein , crura, buccal mucosa, fascia lata	Good in selected patients but out of favour: extra operating time and a second donor site (scarring, lymphoedema). Crural and buccal grafts are too small for large defects
Synthetic	Dacron (polyethylene terephthalate), Teflon (polytetrafluoroethylene)	Not recommended: infection, inflammatory reaction, fibrosis
Allograft and xenograft ("off the shelf")	Tutoplast processed human or bovine pericardium; porcine small intestinal submucosa (SIS) (Surgis ES); porcine skin or dermis	The two most common today are pericardium and SIS. Pericardium is thin and strong, does not contract , and has no reported infection or rejection

Category	Examples	Status and issues
Collagen fleece	TachoSil (collagen fleece coated with sealant)	No suturing needed , shorter operating time, rising popularity. Also used over tunical defects at prosthesis surgery
Tissue-engineered	Adipose stem cell-seeded SIS, acellular matrix, autologous endothelialised tunica	Experimental. Theoretically better take and less fibrosis

- **Pericardium versus dermis:** similar satisfaction. Pericardium had higher recurrence but **shorter operating time and no donor-site morbidity**.
- **SIS: graft contraction** (especially with one-ply grafts) causes recurrent curvature, and sub-graft haematoma and infection occur. Contemporary series are better.
- **Porcine dermis versus bovine pericardium:** similar outcomes, with high rates of ED and perceived shortening.
- **The AUA offers no opinion on graft choice.**
- **Table 76.4 (Outcomes for Plaque Excision or Incision and Grafting):** straightening is generally high across all materials, whereas **ED and satisfaction vary widely**. Dermal, SIS and older synthetic or fleece series are the most variable.

Grafting Surgical Technique

1. **General anaesthesia**, intravenous antibiotics and deep vein thrombosis prophylaxis → **measure dorsal SPL**.
2. **Artificial erection:** vasoactive drug (papaverine, Trimix or prostaglandin E1) through a **21-gauge butterfly needle via the glans** → saline infusion to full rigidity → document curvature, indentation and hinge.
3. **Circumcising incision 1.5–2 cm proximal to the corona** (or through an old circumcision scar) → deglove to Buck fascia → bipolar haemostasis, using **loupe magnification**.
4. **Dorsal or dorsolateral curve:** parallel incisions just lateral to the urethral ridge → elevate **Buck fascia with the neurovascular bundle** off the tunica by sharp dissection, using bipolar cautery for adhesions.
5. Recreate the erection → mark the area of maximum deformity.
 - **Even a pure lateral curve requires the incision or excision to cross the dorsal septum**, the anchor point of the scar, or residual curvature remains.
 - Extensive calcification can be shaved off the inner circular fibres, leaving the outer lamina intact.
6. Create a rectangular defect → **dart the corners radially** → keep the **lateral sides of equal length**, a simplified geometric principle that corrects lateral and dorsal curves. The proximal transverse side is usually longer because the shaft tapers distally. Dorsal length typically gains **0.5–3.0 cm**.
7. Place **4-0 PDS stay sutures** at the four corners and the transverse midpoints → measure the defect on stretch.
 - Size the graft: **pericardium ≤10% larger than the defect; porcine SIS 25% larger**.
 - Secure with running 4-0 PDS. For large defects, add interrupted septal sutures to limit sub-graft blood collection. **TachoSil needs no suturing**.
8. Recreate the erection → **plicate any residual curvature**, which is needed in up to 25%.
 - For a long curve, or an indentation with a distal curve, graft the indentation and plicate the residual curve.
 - A single graft does not carry higher ED than multiple grafts and is faster.

9. Close Buck fascia with running 4-0 chromic → close the skin with interrupted 4-0 chromic horizontal mattress sutures.

- **Circumcision is not mandatory** in uncircumcised men without phimosis. Circumcise if the foreskin is redundant or phimotic, to prevent **paraphimosis**.

10. Dress with Xeroform plus a **Coban wrap applied distal to proximal** → remove at **3 days**, then allow showering but not submersion.

Postoperative Management

- Rehabilitation is as important as after orthopaedic joint replacement: it **reduces ED and length loss and optimises straight healing**.
- **At 2 weeks → massage and stretch**: grasp the glans, stretch, and massage the shaft for **5 minutes twice daily for 2–4 weeks**. The partner can take over in the second fortnight, which helps overcome fear of reinjury.
- **Nightly PDE5I**: vasodilation supports graft take, reduces scar contraction and may preserve cavernosal tissue.
- **Postoperative traction** reduces length loss and can produce **length gain** after both plication and PEG. The regimen is **≥3 hours a day, from 3–4 weeks after surgery, for 3 months**.
- **Grafting outcomes**: straightening 74–100%, postoperative ED 5–54% (with no uniform definition). Long term: ED up to 24% and recurrence 8–12%.
- **AUA**: may offer plaque incision or excision and/or grafting when rigidity is adequate for coitus (Moderate Recommendation; Evidence Grade C).

Penile Prosthesis Surgery

Indications

- **In PD with ED refractory to PDE5Is, the prosthesis is the procedure of choice**: a single operation treats both problems.
- **Adjunctive straightening**: manual modelling, tunical incision ± graft, or **transcorporal** incision or stretch of the plaque from within the corpus.
- **An inflatable penile prosthesis (IPP) is preferred**: cylinder pressure allows better correction with modelling and better girth. **Malleable devices gave narrow, cold, unnatural erections**.

Surgical Technique and Adjunctive Procedures for Penile Straightening

- **Approach**: penoscrotal, with a high-pressure inflatable cylinder. All three-piece and two-piece devices have been used successfully.

Manual modelling (Wilson and Delk, 1994) — the first-line step:

1. Place the cylinders → close the corporotomies → fill to full rigidity through a **surrogate reservoir**.
2. **Clamp the tubing between pump and cylinders with shodded haemostats** to protect the pump from high pressure.
3. Bend opposite to the curve → **hold 60–90 seconds** (in practice often ≤30 seconds) → reinflate and repeat until straight.
4. Bend **gradually, not violently**, to avoid tunical tears and neurovascular injury.
5. **Place the bending hand on the shaft, not the glans**. Pressure on the glans drives the cylinder tips through the **fossa navicularis** and causes urethral injury.

6. The other hand supports the base **over the corporotomies** to protect the suture lines.

- **Modelling outcomes:** straightening in 86–100%, with **no increase in device revision**. Sensory deficit is rare but must be discussed. **Modelling is appropriate as first-line therapy even for severe curves.**
- **Alternative:** plication contralateral to the curve **before** prosthesis placement. For a dorsal curve, define the deformity with vasoactive drug and saline infusion, then place Lembert sutures ventrally.
- **Residual curve >30° after modelling, or indentation causing cylinder buckling** → elevate Buck fascia → make a **relaxing tunical incision** over the maximum curve.
 - Incise with the cylinders deflated, preserving cavernosal tissue over the implant.
 - Use cautery **<30 W with Titan cylinders** to avoid thermal injury.
 - Reinflate and remodel.
- **Graft if the defect is >2 cm** in any dimension, to prevent scar contracture and **cylinder herniation**. Use pericardium or SIS; **never locally harvested dermis**, which risks transferring bacteria to the prosthesis. Collagen fleece has also been used.
- **Home modelling**, for residual curvature <45° after IPP placement and modelling: start at **4 weeks** → fully inflate → bend opposite the curve and hold 30 seconds → deflate. Continue cycles for 20 minutes, 3 times daily, for 6 months. **Most patients reach ≤10°.**
- A **transverse penoscrotal incision** reaches almost the whole shaft, so **degloving is rarely needed**, except for a distal dorsal curve.
- **Sliding technique** (with a prosthesis): restores length and girth in debilitating length or girth loss, with mean gains of about 3 cm. Most patients received malleable devices, all curves were corrected, and experience is from a single institution.
- **Long-term IPP outcomes in PD:**
 - Most patients needed modelling; a minority needed incision ± pericardial graft.
 - Additional maneuvers did **not increase mechanical failure or infection**.
 - Satisfaction with curvature was lower than overall satisfaction, which underlines the need for **expectation-setting**.
- **Functional straightness:** a residual curve of ≤20° is acceptable and may improve with time through cylinder-driven tissue expansion.
- **The two North American three-piece devices do not differ** in reliability, infection or satisfaction.
- **Table 76.6 (Outcomes of Penile Prosthesis Implantation for Peyronie Disease):** most series used inflatable devices. The need for additional straightening maneuvers varied widely, and satisfaction was generally good.

Complications

Length loss.

- **Length loss is the commonest complaint after a prosthesis.** Objective loss is under 1 cm by 1 year, which is especially distressing in PD.
- **Pre-prosthesis traction** (3–4 months at ≥3 hours a day) resulted in **no further length loss** after implantation, and most patients gained length.
- **Postoperative prolonged inflation:** 10–30 minutes a day for 3 months, starting 6 weeks after surgery. This maintains length and reduces residual curvature.
- With the **sliding technique**, **subcoronal degloving carries a risk of ischaemia or necrosis**. A **non-degloving ventral incision** preserves the skin, dartos and neurovascular bundle and has avoided vascular complications.

KEY POINTS — SURGICAL MANAGEMENT

- **Surgery (± prosthesis) is the gold standard** when deformity or rigidity compromises or prevents penetration.
- **Thorough consent** about limitations improves satisfaction.
- **Satisfactory rigidity plus a curve <60–70° without significant indentation → plication.** No technique is proven superior, as there are no head-to-head trials.
- **Severe or complex deformity plus strong erections plus no venous insufficiency on PDDUS → PIG/PEG.**
- **Complications of all straightening operations:** incomplete straightening, recurrence, shortening, reduced sensation and ED.
- **Outcome depends on patient selection and technique, not graft type.**
- **Inadequate rigidity → prosthesis with straightening maneuvers as first-line surgery.**
- **AUA 2015:**
 - Prosthesis for PD with ED and/or deformity preventing coitus despite pharmacotherapy or VED (Moderate; Grade C).
 - Adjunctive modelling, plication or incision/grafting for persistent deformity (Moderate; Grade C).
 - **Use an inflatable device.**

CONCLUSION

- PD is **far more common than previously thought** and is a growth area for both clinical practice and research.
- Recognise the **acute and stable phases**, and **operate only once the scar and deformity are stable** (≥12 months from onset, stable ≥6 months).
- **Complete correction** of curvature, indentation and shortening **is unlikely**; the **goal is sexual function**.
- Recognise the **devastating psychological impact**, and offer counselling where indicated.
- In the acute phase, the **only guideline-endorsed oral therapy is NSAIDs for pain**.
- **Surgical risks:** incomplete straightening, recurrence, further shortening, altered sensation and, most importantly, **diminished rigidity**.
- **Drug-refractory ED plus PD → prosthesis with straightening maneuvers**, which addresses both in one operation.